

Outline

Review: Event Handling

Event Handling Programming via Greenfoot

Determining Event Sources

Managing Fonts

Check Boxes and Radio Buttons

Event Handling Programming

- The BlueJ team also made Greenfoot
 - Download: <https://www.greenfoot.org/download>
- Greenfoot is a simple game engine for novice Java programmers
- Sample Greenfoot game:
 - Hedgehog eats cheese, dodges balloons
 - <https://www.greenfoot.org/scenarios/20575>

state and act() callback method

```
int speed = 2;  
int Lives = 3;  
int Points = 0;  
  
public void act()  
{  
    moveWithKeys();  
    showStatus();  
    hitPizza();  
    hitBalloon();  
    testEndGame();  
    testWinGame();  
}
```

The code to move the Hedgehog

```
public void moveWithKeys()
{
    if (Greenfoot.isKeyDown("right") )
    {
        setLocation(getX()+speed, getY() );
    }
    if (Greenfoot.isKeyDown("Left") )
    {
        setLocation(getX()-speed, getY() );
    }
    if (Greenfoot.isKeyDown("Up") )
    {
        setLocation(getX(), getY()-speed );
    }
    if (Greenfoot.isKeyDown("Down") )
    {
        setLocation(getX(), getY()+speed );
    }
}
```

Collisions with pizza and balloons

```
public void hitPizza()  
{  
    if (isTouching (pizza.class) )  
    {  
        removeTouching (pizza.class);  
        addPoints();  
    }  
}
```

```
public void hitBalloon()  
{  
    if (isTouching (balloon.class) )  
    {  
        setLocation(300,100);  
        loseLife();  
    }  
}
```

Win and lose conditions

```
public void testEndGame()  
{  
    if (Lives<1)  
    {  
        getWorld().showText("YOU LOSE", 300, 200);  
    }  
}
```

```
public void testWinGame()  
{  
    if (Points>3)  
    {  
        getWorld().showText("YOU WIN", 300, 150);  
    }  
}
```

All of the code for the balloon

```
import greenfoot.*;  // (World, Actor, GreenfootImage)

/**
 * Write a description of class balloon here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class balloon extends Actor
{
    /**
     * Act - do whatever the balloon wants to do. This
     * the 'Act' or 'Run' button gets pressed in the
     */
    public void act()
    {
        move(1);
        if (isAtEdge() )
        {
            turn(180);
            move(1);
        }
    }
}
```

Status information

```
public void showStatus()  
{  
    getWorld().showText("Lives"+Lives, 50, 30);  
    getWorld().showText("Points"+Points, 50, 10);  
}
```

```
public void addPoints()  
{  
    Points = Points + 1;  
}
```

```
public void loseLife()  
{  
    Lives = Lives - 1;  
}
```